LAB PROGRAM-2

2.Develop a Java program to create a class Book with members Book ID, Title, Author, and Price. Use a constructor to initialize the details. Include methods to display book information, overload search() to search by Book ID and Title, and create a method that accepts another Book object and returns the costlier book. Use a static member to maintain the total number of books created. In the main class, create multiple book objects and demonstrate all the Operations.

```
import java.util.Scanner;
class Book {
    int bookId;
    String title;
    String author;
    double price;

    static int totalBooks = 0;

   Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
        totalBooks++;
    }

    void display() {
        System.out.println("Book ID: " + bookId);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
        System.out.println();
    }

    void search(int id) {
        if (bookId == id)
            display();
        else
            System.out.println("Book ID not found");
    }

    void search(String title) {
        if (this.title.equalsIgnoreCase(title))
            display();
```

```
        else
            System.out.println("Book Title not found");
    }

    Book costlierBook(Book b) {
        if (this.price > b.price)
            return this;
        else
            return b;
    }

    static void displayTotalBooks() {
        System.out.println("Total Books Created: " + totalBooks);
    }
}

public class Main {
    public static void main(String[] args) {
        Book b1 = new Book(100, "Java Programming", "James Gosling", 400);
        Book b2 = new Book(101, "Discrete mathematics", "Mark Allen", 680);
        Book b3 = new Book(102, "DLCO", "Raghu Ramakrishnan", 750);

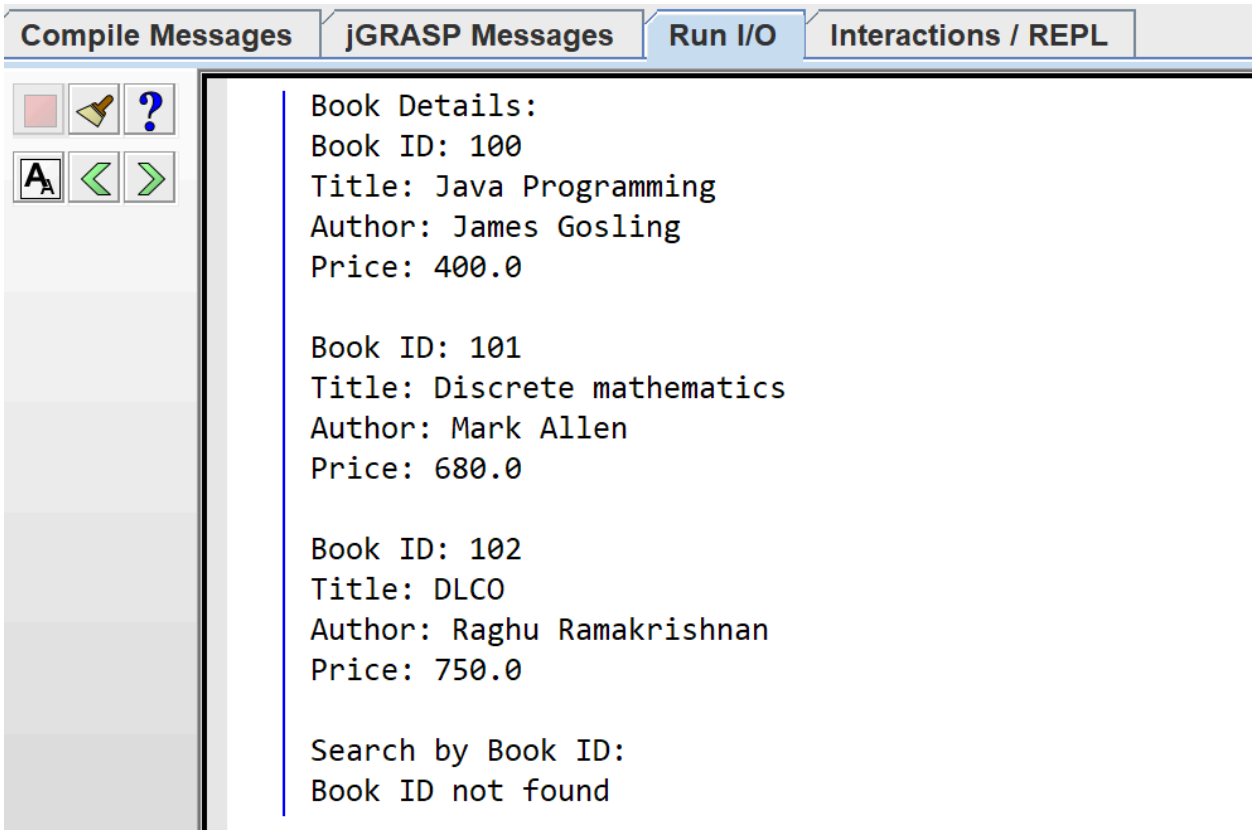
        System.out.println("Book Details:");
        b1.display();
        b2.display();
        b3.display();

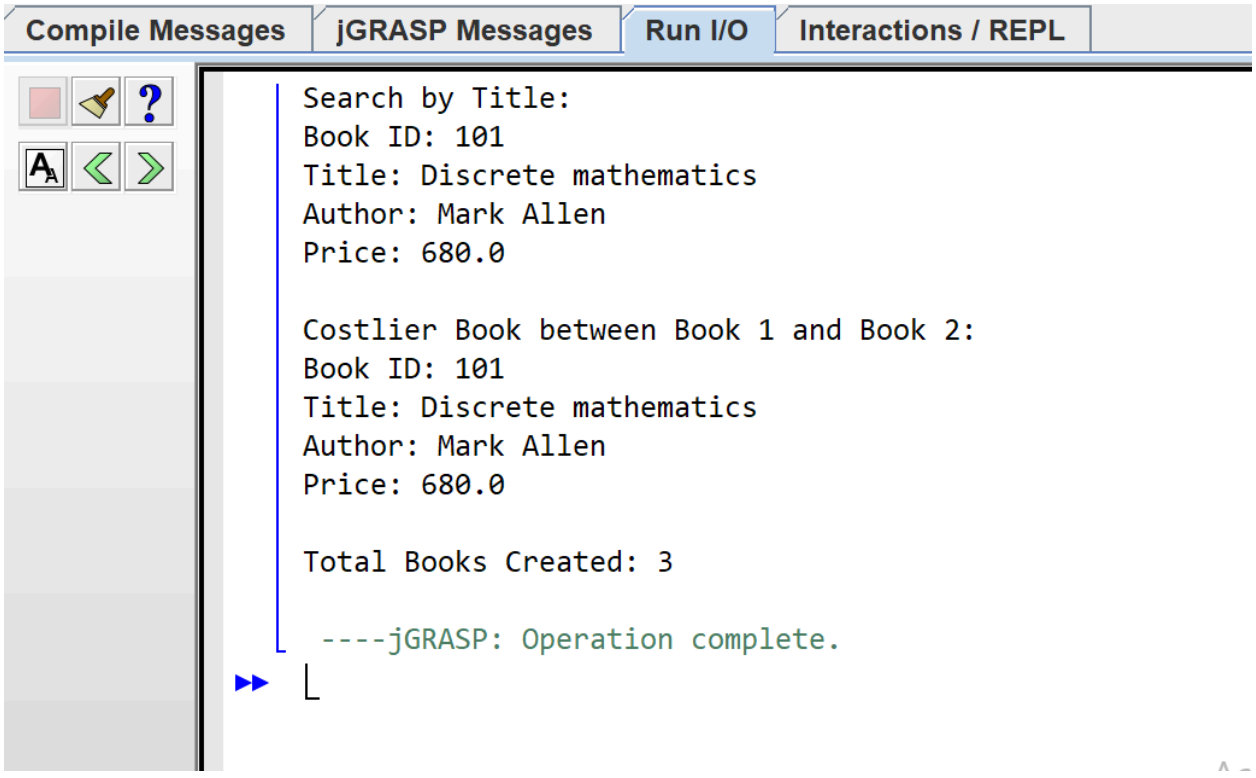
        System.out.println("Search by Book ID:");
        b1.search(101);

        System.out.println("Search by Title:");
        b2.search("Discrete mathematics");

        System.out.println("Costlier Book between Book 1 and Book 2:");
        Book costly = b1.costlierBook(b2);
        costly.display();

        Book.displayTotalBooks();
    }
}
```


Compile Messages
jGRASP Messages
Run I/O
Interactions / REPL
Book Details:
Book ID: 100
Title: Java Programming
Author: James Gosling
Price: 400.0
Book ID: 101
Title: Discrete mathematics
Author: Mark Allen
Price: 680.0
Book ID: 102
Title: DLCO
Author: Raghu Ramakrishnan
Price: 750.0
Search by Book ID:
Book ID not found


Compile Messages
jGRASP Messages
Run I/O
Interactions / REPL
Search by Title:
Book ID: 101
Title: Discrete mathematics
Author: Mark Allen
Price: 680.0
Costlier Book between Book 1 and Book 2:
Book ID: 101
Title: Discrete mathematics
Author: Mark Allen
Price: 680.0
Total Books Created: 3
----jGRASP: Operation complete.